

Linear Algebra

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Chapter 0

Prerequisites

0.1 Sets

Lecture 01: Sets

Definition (Set). Collection of objects where objects can be almost anything (number, symbol, set, shape...)

Note. This definition can lead to paradoxes, but its fine. Axiomatic set theories avoid this (e.g., Zermelo-Fraenkel), but usually the subtleties are not necessary.

- A set is described by the objects that **belong** to it (are **in** it)
- Sets are given (usually single, upper-cased, italicized, roman letter) names: A, B, X, P, R, T
- An object in a set is a **member** or **element** of A
- Belonging to (being a member of, being in) a set is denoted by $\in$ (e.g., $2 \in A$)

Definition 0.1.1 (Equal). Two sets, A and B are **equal** (denoted $A = B$) if every member of A is also a member of B and every member of B is also a member of A

Note. There is no order of members in a set (no “first,” or “last” member)

0.1.1 Set-builder notation

- To describe a small set, we can list members explicitly with curly braces, separated by commas (e.g., $A = \ominus, \ominus, \star$)
- For larger (possibly infinite) sets we describe members using a predicate
 - A is the set of students in Quiz Bowl club
 - $\mathbb{N}$ is the set of Natural numbers
- The set-builder notation, $\{x : \Phi(x)\}$, is a concise expression of this
 - $A = \{x : x \text{ is a student in Quiz Bowl club}\}$
 - $B = \{x : x^2 = 4\}$
 - $C = \{2k : k \in \mathbb{N}\}$

Definition (Predicate). A logical formula that evaluates to True ($\top$) or False ($\perp$)

Definition (Domain of Discourse). Universe of objects that can potentially be in the set if they satisfy the predicate

Usually implied from the context, but can be explicitly defined:

$$E \in \mathbb{N} : (x \% 2) = 0\}$$

where $\mathbb{N}$ is the set of natural numbers (counting numbers):

$$\mathbb{N} = 1, 2, 3, \dots$$

0.1.2 Logic

Definition (Conditional Operator). Denoted $p \Rightarrow q$ (“if p then q ” or “ p implies q ”)

$$(p \Rightarrow q) = \begin{cases} \perp & \text{if } p = \top, q = \perp \\ \top & \text{otherwise} \end{cases}$$

Definition (Conjunction Operator). Denoted $p \wedge q$ (“ p and q ”)

$$(p \wedge q) = \begin{cases} \top & \text{if } p = \top, q = \top \\ \perp & \text{otherwise} \end{cases}$$

Definition (Disjunction Operator). Denoted $p \vee q$ (“ p or q ”)

$$(p \vee q) = \begin{cases} \perp & \text{if } p = \perp, q = \perp \\ \top & \text{otherwise} \end{cases}$$

Definition (Negation Operator). Denoted $\neg p$ (“not p ”)

$$(\neg p) = \begin{cases} \top & p = \perp \\ \perp & p = \top \end{cases}$$

Definition (Biconditional Operator). Denoted $p \Leftrightarrow q$ (“ p if and only if q ”)

$$(p \Leftrightarrow q) = (p \Rightarrow q) \wedge (q \Rightarrow p)$$

0.1.3 Set Notation and Terminology

Let A, B be sets from the same [domain of discourse](#)

Definition 0.1.2 (Subset). A is called a **subset** of B , denoted $A \subseteq B$, if

$$(x \in A) \Rightarrow (x \in B)$$

Note.

$$A \subseteq A$$

Definition 0.1.3 (Intersection). The **intersection** of A and B , denoted $A \cap B$, is the set

$$(A \cap B) = \{x : (x \in A) \wedge (x \in B)\}$$

Definition 0.1.4 (Union). The **union** of A and B , denoted $A \cup B$, is the set

$$(A \cup B) = \{x : (x \in A) \vee (x \in B)\}$$

Notation. The **empty set**, denoted $\emptyset$, contains no members

Note.

$$\emptyset \subseteq A$$

Definition 0.1.5 (Power Set). Let A be a set; the **power set of** A , denoted $P(A)$, is the set of all subsets of A :

$$P(A) = \{S : S \subseteq A\}$$

Example.

$$A = \{\ominus, \ominus, \star\}$$
$$P(A) = \{\emptyset, \{\ominus\}, \{\ominus\}, \{\star\}, \{\ominus, \ominus\}, \{\ominus, \star\}, \{\ominus, \star\}, \{\ominus, \ominus, \star\}\}$$

Definition 0.1.6 (Cartesian Product). Let A, B be sets; the **Cartesian product** of A with B , denoted $A \times B$ is the set of all ordered pairs of items, the first taken from A and the second taken from B

$$A \times B = \{(x, y) : x \in A, y \in B\}$$

Example.

$$A = \{\ominus, \ominus, \star\}, B = \{\circ, \diamond\}$$
$$A \times B = \{(\ominus, \circ), (\ominus, \diamond), (\ominus, \circ), (\ominus, \diamond), (\star, \circ), (\star, \diamond)\}$$

Note. We denote an ordered pair with parenthesis, not curly braces

- (x, y) is not the same as x, y because order matters
- $x, y = y, x$ but $(x, y) \neq (y, x)$

0.2 Proofs

Lecture 02: Proofs

0.2.1 Deductive Reasoning

Definition (Deductive Reasoning). The process of making deductive arguments

Definition (Deductive Argument). Process of making a logical inference

Definition (Inference). Claim that a certain [predicate](#), called the **conclusion**, follows from one or more [predicates](#), called the **premises**.

Predicate B **follows from** A if it is impossible simultaneously for A to be True and B to be False

An inference is **valid** if the conclusion follows from the premises

A deductive argument is **sound** if the inference is valid and its premises are True

0.2.2 More Logic Notation and Terminology

Definition (Universal Quantification Symbol ($\forall$)). Denotes that a proposition is True for all members

Example.

$$\forall x \in \mathbb{N} : x^2 \geq x$$

Read: “**for all** (every, any, each) x in $\mathbb{N}$ the predicate $(x^2 \geq x)$ evaluates to True”

Definition (Existential Quantification Symbol($\exists$)). Denotes that a proposition is True for at least one member

Example.

$$\exists x \in \mathbb{N} : x^2 < x$$

Read: “**there exists** an x in $\mathbb{N}$ such that the predicate $x^2 < x$ evaluates to true”

Note. This is an example of a False proposition

0.2.3 Contraposition

Definition (Contraposition). Let p, q be predicates and consider the conditional statement

$$p \Rightarrow q$$

The **contrapositive** form of the statement is

$$\neg q \Rightarrow \neg p$$

A conditional statement and its contrapositive form are **equivalent**

$$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$$

When judging truthfulness of a statement, it sometimes helps to consider its contrapositive

Also existing, but not as common are:

- The **inverse**: $\neg p \Rightarrow \neg q$
- The **converse**: $q \Rightarrow p$
- The **complement**: $\neg(p \Rightarrow q)$

0.2.4 Proofs

Definition (Mathematical Proof). A deductive argument about something related to math. Uses spoken/written language, or even sketches/diagrams.

Usually rigorous (spells out assumptions and deductive steps as is convenient) but informal (some natural language with occasionally ambiguous symbols/rules) deductive reasoning

Proposition 0.2.1. Let $n, m \in \mathbb{N}$ and suppose n, m are even; then $(n + m)$ is even

Proof.

- Since n is even $\exists k \in \mathbb{N}$ such that $n = 2k$
- Since m is even $\exists q \in \mathbb{N}$ such that $m = 2q$
- Then $(n + m) = (2k + 2q) = 2(k + q)$
- Therefore, $(n + m)$ is even

Note. We used a method called **direct proof**

0.2.5 Mathematical Induction

Definition (Mathematical Induction). If asked to prove that a certain proposition, $P(n)$ is true for any $n \in \mathbb{N}$, we will accept as proof the following inference, if sound:

$$\left(\begin{array}{l} P(1) = T \\ P(k) \stackrel{2}{=} T \end{array} \Rightarrow (P(k+1) = T) \right) \stackrel{3}{\Rightarrow} P(n) = T, \forall n \in \mathbb{N} \quad (1)$$

Where (1) is called the **base case**; (2) is called the **induction hypothesis**; (3) is the **induction step**

A variant called **complete** (or strong/generalized) induction uses a stronger hypothesis in the induction step:

$$(P(j), \forall j \leq k) \Rightarrow P(k+1)$$

Proposition 0.2.2. For any $n \in \mathbb{N}$

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Proof. By induction

- Base case: for $n = 1$, $\frac{n(n+1)}{2} = 1$
- Induction step: suppose, for some $k \in \mathbb{N}$, $1 + 2 + \dots + k = \frac{k(k+1)}{2}$, then

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2} \end{aligned}$$

Proposition 0.2.3. Let $n \in \mathbb{N}$, if n^2 is even then n is even

Proof. By Contraposition

Suppose n were not even, then

- n is odd, therefore
- $\exists k \in \mathbb{N}$ such that $n = 2k - 1$, and therefore
- $n^2 = n \cdot n = (2k - 1)(2k - 1) = 4k^2 - 4k + 1 = 4(k^2 - k) + 1$
- Let $p = 4(k^2 - k)$, then
- $p \div 2 = 2(k^2 - k)$, therefore
- p is even, therefore
- $n^2 = p + 1$ is odd
- Since n^2 is not odd, n cannot be not even

Proposition 0.2.4. $\sqrt{2}$ is irrational

Proof. By contradiction

Suppose $\sqrt{2}$ is rational, then

- $\exists p, q \in \mathbb{N}$ such that $\sqrt{2} = \frac{p}{q}$ and p and q have no common factors,

0.2.7 Proof of Equivalence

To prove $P \Leftrightarrow Q$, prove both $P \Rightarrow Q$ and $Q \Rightarrow P$ Related: show two sets A, B are equal by showing $A \subseteq B$ and $B \subseteq A$ To prove the following are equivalent (TFAE) you could prove a circular chain of implications

$$(A \Rightarrow B) \wedge (B \Rightarrow C) \wedge (C \Rightarrow D) \wedge (D \Rightarrow A)$$

0.3 Operations

Lecture 03: Operations

0.3.1 Function

Definition 0.3.1 (Function). A function comprises three objects:

- A set, called the **domain**
- Another set, called the **range**
- A mapping of each member of the domain to a single member of the range, called its **image**

Notation. Often we use lower-case Latin letters to name functions.

The notation $f : D \rightarrow R$ reads “ f is a function from the domain set D into the range set R ”.

The image of $x \in D$ is denoted $f(x)$ (read “ f of x ”).

Definition 0.3.2 (Image of a Set). Let $f : D \rightarrow R$ be a function and let $U \subseteq D$.

The set of images of members of U is called the **image of U** , and denoted $f(U)$:

$$f(U) = \{f(x) : x \in U\}$$

The set of images of all members of the domain, $f(D)$, is called the **image of the function**.

Example. Let $D = \{-2, 2, 3, 5, 7\}$ and let $h : D \rightarrow \mathbb{N}$ be the mapping

$$h(-2) = 4, \quad h(2) = 4, \quad h(3) = 9, \quad h(5) = 25, \quad h(7) = 49$$

The domain and range are given explicitly. The image is evident from the mapping; the image of h is the set

$$S = \{4, 9, 25, 49\}$$

With D and S as above, let $f : D \rightarrow S$ be the mapping

$$f(-2) = 4, \quad f(2) = 4, \quad f(3) = 9, \quad f(5) = 25, \quad f(7) = 49$$

Strictly speaking, $f : D \rightarrow S$ and $h : D \rightarrow \mathbb{N}$ are different functions (they have different ranges). This distinction is not that important.

0.3.2 Ways to Describe a Function

There are many different ways to describe a function:

- **A table:**

Domain	-2	2	3	5	7
Image	4	4	9	25	49

- **A formula:** $f(x) = x^2$
- **A set of ordered pairs:** $f = \{(-2, 4), (2, 4), (3, 9), (5, 25), (7, 49)\}$
- **A set of ordered pairs in set builder notation:**

$$f = \{(x, y) : x \in \{-2, 2, 3, 5, 7\}, y = x^2\}$$

Note. Not every set of ordered pairs describes a valid function. For example:

$$\{(2, 2), (2, 3), (3, 1)\}$$

is not a function because the element 2 in the domain maps to two different values.

0.3.3 Bijection

Definition 0.3.3 (Injective, Surjective, Bijective). Let $f : D \rightarrow R$ be a function.

- f is called **injective** (or **one-to-one**) if distinct members of D are mapped to distinct members of R :

$$\forall x, x' \in D, \quad x \neq x' \Rightarrow f(x) \neq f(x')$$

- f is called **surjective** (or **onto**) if every element of R is the image of some element of D :

$$\forall y \in R, \quad \exists x \in D : y = f(x)$$

- f is called **bijective** if it is both injective and surjective (one-to-one and onto).

Note. Students are sometimes confused about the surjective part because, in calculus, it is customary to let the range of a function be implicitly defined as equal to its image, making every function surjective.

0.3.4 Binary Operations

Remember the four basic arithmetic operations? What they have in common is that they each take two numbers as input and produce one number as output.

Definition 0.3.4 (Binary Operation). Let A be a set. A **binary operation** on A is a function whose domain is the **Cartesian Product** $A \times A$.

Notation. An operation is often denoted with a symbol, like $\star$, instead of a letter, and the image of a pair from the domain is denoted with the operator

0.2.6 Things that often need proving

- Existence: $\exists x$ such that $P(x)$ Proof by construction describes who the required x is, and verifies $P(x) = T$ (Don't need to actually find the thing)
- Uniqueness: $(P(x) \wedge P(y)) \Rightarrow (x = y)$ Name two things with the property and show they must be equal

between the pair items:

$$x \star y \text{ instead of } f(x, y)$$

Example. Some examples of binary operations $\star$ on $\mathbb{N}$ are:

- $a \star b := a + b + 8$
- $a \star b := \max(a, b)$
- $a \star b :=$ the digit in the ones place of $a + b^2$

0.3.5 Range of Operations and Closure

The sum and product of any two natural numbers is a natural number:

$$\begin{aligned} \forall a, b \in \mathbb{N}, \quad a + b \in \mathbb{N} \\ \forall a, b \in \mathbb{N}, \quad ab \in \mathbb{N} \end{aligned}$$

This is not true for subtraction and division:

$$\begin{aligned} \exists a, b \in \mathbb{N} : a - b \notin \mathbb{N} \\ \exists a, b \in \mathbb{N} : a/b \notin \mathbb{N} \end{aligned}$$

This convenient property of addition and multiplication is called **closure**.

Definition 0.3.5 (Closure). A set A is said to be **closed with respect to an operation $\star$** if

$$\forall a, b \in A, \quad (a \star b) \in A$$

Example.

- The set $\mathbb{N}$ is closed with respect to addition and with respect to multiplication
- The set $\mathbb{N}$ is not closed with respect to subtraction
- The set $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is closed with respect to subtraction

Definition 0.3.6 (Associative Operation). An operation $\star$ on a set A is called **associative** if, $\forall a, b, c \in A$:

$$(a \star b) \star c = a \star (b \star c)$$

Example.

- Addition and multiplication are associative operations on $\mathbb{Z}$
- Subtraction is not associative on $\mathbb{Z}$

Note. If $\star$ is an associative operation on A , it is customary to neglect parentheses:

$$(a \star b) \star c = a \star (b \star c) = a \star b \star c$$

It is implied in the definition of associativity that the set A is closed with respect to $\star$.

Problem (CFU). If $\star$ is associative on A , is it true that $\forall a, b, c, d \in A$:

$$(a \star b) \star (c \star d) = a \star (b \star c) \star d$$

Bonus CFU: can you prove your answer?

Answer. Yes.

$$(a \star b) \star (c \star d) = ((a \star b) \star c) \star d = (a \star (b \star c)) \star d = a \star (b \star c) \star d$$

⊗

Theorem 0.3.7 (Generalized Associativity). Let $\star$ be an associative operation on a set F . Then, for any $a_1, a_2, \dots, a_n \in F$, every possible parenthesis ordering of

$$a_1 \star a_2 \star a_3 \star \dots \star a_{n-1} \star a_n$$

is equivalent to the left-associated ordering:

$$(\dots((a_1 \star a_2) \star a_3) \star \dots \star a_n)$$

Proof. By generalized (strong) induction.

Base case: For $n = 1$ or $n = 2$ there is nothing to prove, and for $n = 3$, there are two possible orderings, and regular associativity guarantees that $(a_1 \star a_2) \star a_3$ is equal to the left-associated ordering $((a_1 \star a_2) \star a_3)$.

Inductive step: For $n > 3$, we suppose that every parenthesis ordering of an expression with fewer than n operands is equivalent to the left-associated ordering, and proceed to consider the expression with n operands:

$$E = a_1 \star a_2 \star \dots \star a_n$$

Any ordering of parentheses ends with a final step:

$$E = L \star R$$

where L is some parenthesis ordering of $a_1 \star \dots \star a_q$ and R is some parenthesis ordering of $a_{q+1} \star \dots \star a_n$, and $q < n$.

By the induction hypothesis, both L and R are equal to their respective left-associated orderings:

$$\begin{aligned} L &= (\dots((a_1 \star a_2) \star \dots) \star a_q) \\ R &= (\dots((a_{q+1} \star a_{q+2}) \star \dots) \star a_n) \end{aligned}$$

If $q = n - 1$, so that $R = a_n$, then $E = L \star R$ is already left-associated. Otherwise, write

$$R = M \star a_n = (\dots((a_{q+1} \star a_{q+2}) \star \dots) \star a_{n-1}) \star a_n$$

and

$$E = L \star (M \star a_n) = (L \star M) \star a_n$$

by regular associativity. By the induction hypothesis, since $L \star M$ includes $n - 1$ operands, it is equivalent to the left-associated ordering of $a_1 \star \dots \star a_{n-1}$, making $E = (L \star M) \star a_n$ the left-associated ordering of the original expression. ■

0.3.7 Neutrality

Definition 0.3.8 (Neutral Element). Let $\star$ be an operation on the set A . An element $e \in A$ is called **neutral with respect to $\star$** if

$$\forall a \in A, \quad a \star e = e \star a = a$$

Theorem 0.3.9 (Uniqueness of the Neutral). Let $\star$ be an operation on a set A . There is at most one element in A that is neutral with respect to $\star$.

Proof. Suppose $p, q \in A$ are both neutral with respect to $\star$. In that case:

- $p \star q = q$, because p is neutral
- $p \star q = p$, because q is neutral
- Therefore, $p = q$

■

Note. Since a neutral element in A is unique, if it exists, we can call it **the** neutral.

Problem (Exercise). Let $\star$ denote the operation on $\mathbb{N}$ defined by $a \star b := a^b$.

- Is there, in $\mathbb{N}$, a neutral with respect to $\star$?
- Can you prove your answer?

Answer. (i) No

(ii) Yes:

- If $a^x = a$, $\forall a \in \mathbb{N}$, then in particular $2^x = 2$ and therefore $x = 1$
- So, the only candidate for being a neutral with respect to $\star$ is 1
- However, if 1 is neutral with respect to $\star$, then $1^a = a$, $\forall a \in \mathbb{N}$, and in particular $1^2 = 2$
- Which it is not
- So there is no neutral element

⊗

0.3.8 Invertible Elements

Definition 0.3.10 (Invertible Element). Let $\star$ be an operation on A and let $e \in A$ be the neutral with respect to $\star$. An element $a \in A$ is called **invertible** if there exists an element $b \in A$ such that

$$\begin{aligned} a \star b &= e \\ b \star a &= e \end{aligned}$$

If such an element b exists, it is called **an inverse of a** .

Example.

- Every element of $\mathbb{Z}$ is invertible with respect to addition
- In $\mathbb{N}$, 1 is the only invertible element with respect to multiplication
- In $\mathbb{Z}$, 1 and -1 are the only invertible elements with respect to multiplication

Chapter 1

Fields

1.1 Definition

Lecture 04: Definition

1.1.1 Algebra

Algebra is the study of equations and calculations. Calculations are done by carrying out operations on objects of an **algebraic structure**.

Definition (Algebraic Structure). A set of objects with operations defined on it.

1.1.2 Numbers as Models

Numbers are an algebraic structure constructed to model certain types of physical objects and their interactions:

- Objects such as apples and skittles
- Interactions such as trading skittles for apples

Interactions in the natural world have certain characteristics, discovered by observation. Operations on numbers have properties matching those observations.

1.1.3 The Natural Numbers Model

The natural numbers $\mathbb{N}$ consist of:

- A set $\mathbb{N}$ of objects, called numbers
- An algorithm assigning each number a unique name
- An algorithm for deciding which number is "next"
- A binary operation called addition, denoted $+$
- An algorithm for finding $a + b$ for any $a, b \in \mathbb{N}$

These algorithms are constructed to have properties modeled after observed characteristics of combining physical objects.

1.1.4 The Integer Numbers Model

Additional observations reveal interactions not adequately modeled by $\mathbb{N}$. The integers $\mathbb{Z}$ extend $\mathbb{N}$ by:

- Adding a new number, called zero and denoted 0
- Matching each number $a \in \mathbb{N}$ with a new number $-a$
- Extending the addition algorithm for these new numbers

1.1.5 Abstraction and Generalization

Exploration in mathematics proceeds by abstraction and generalization:

1. Start with a familiar mathematical model whose properties are understood
2. Investigate a hypothetical, abstract structure defined by those properties
3. Any claim proven about the abstract structure is true for any concrete model satisfying the same axioms

1.1.6 Groups

Definition 1.1.1 (Group). A group is a set G equipped with an operation $\star$ that satisfies:

- (i) $\forall a, b \in G, a \star b \in G$ (closure)
- (ii) $\forall a, b, c \in G, (a \star b) \star c = a \star (b \star c)$ (associativity)
- (iii) $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$ (neutral element)
- (iv) $\forall a \in G, \exists b \in G$ such that $a \star b = b \star a = e$ (invertibility)

Definition 1.1.2 (Commutative Group). A commutative group is a group that also satisfies $\forall a, b \in G, a \star b = b \star a$.

Example.

- $(\mathbb{Z}, +)$ is a group
- $(\mathbb{Z}, \cdot)$ is not a group (fails invertibility)
- Hours on a clock with "passage of time" operation forms a group

Groups are important because many different things can be modeled as groups, but they don't capture everything we need for linear algebra.

1.1.7 The Rational Numbers Model

Additional observations lead to the need for division. The rational numbers are:

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$$

We can represent rational numbers as ordered pairs (a, b) where:

- Equality: $(a, b) = (c, d) \Leftrightarrow ad = bc$
- Addition: $(a, b) + (c, d) = (ad + bc, bd)$
- Multiplication: $(a, b) \cdot (c, d) = (ac, bd)$

1.1.8 Properties of $\mathbb{Q}$

From the definition of operations on $\mathbb{Q}$:

- (i) $\mathbb{Q}$ is closed with respect to both addition and multiplication
- (ii) Both operations are associative
- (iii) Both operations are commutative
- (iv) $(0, 1) \in \mathbb{Q}$ is neutral with respect to addition, $(1, 1) \in \mathbb{Q}$ is neutral with respect to multiplication
- (v) Multiplication is distributive over addition: $a(b + c) = ab + ac$
- (vi) Every $q \in \mathbb{Q}$ is invertible with respect to addition; every $q \neq (0, 1)$ is invertible with respect to multiplication

1.1.9 Fields

Definition 1.1.3 (Field). A field is a set F equipped with two binary operations, called addition and multiplication and denoted $+$ and $\cdot$ respectively, that satisfy:

- (i) $\forall a, b \in F: a + b \in F$ and $a \cdot b \in F$ (closure)
- (ii) $\forall a, b, c \in F: (a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ (associativity)
- (iii) $\forall a, b \in F: a + b = b + a$ and $a \cdot b = b \cdot a$ (commutativity)
- (iv) $\forall a, b, c \in F: a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ (distributivity)
- (v) $\exists \tilde{0} \in F$ and $\exists \tilde{1} \in F$ such that $\forall a \in F: a + \tilde{0} = a$ and $a \cdot \tilde{1} = a$ (neutral elements)
- (vi) $\forall a \in F, \exists a' \in F$ such that $a + a' = \tilde{0}$ (additive inverses)
- (vii) $\forall a \neq \tilde{0} \in F, \exists a' \in F$ such that $a \cdot a' = \tilde{1}$ (multiplicative inverses)
- (viii) $\tilde{0} \neq \tilde{1}$ (distinct neutral elements)

Note. The two operations are not completely symmetric

- Multiplication distributes over addition, but not vice versa
- The additive neutral is not required to have a multiplicative inverse

1.1.10 Notation and Terminology

- Elements of a field are called **scalars**
- We drop the tildes and refer to $\tilde{0}$ and $\tilde{1}$ as "zero" and "one"
- Multiplication precedes addition in order of operations
- We often omit the multiplication symbol: $ab + ac$ instead of $a \cdot b + a \cdot c$

1.1.11 Uniqueness of Inverses

Proposition 1.1.4 (Uniqueness of Additive Inverse). Let F be a field and $a \in F$. There exists a unique $a' \in F$ such that $a + a' = 0$.

Proof. Let $a \in F$ and suppose $a', a'' \in F$ such that

$$\begin{aligned} a' + a &= a + a' = 0 \\ a'' + a &= a + a'' = 0 \end{aligned}$$

Then:

$$a' = a' + 0 = a' + (a + a'') = (a' + a) + a'' = 0 + a'' = a''$$

Proposition 1.1.5 (Uniqueness of Multiplicative Inverse). Let F be a field and $a \neq 0 \in F$. There exists a unique $a' \in F$ such that $a \cdot a' = 1$.

Proof. Let $a \in F$ and suppose $a', a'' \in F$ such that

$$\begin{aligned} a' \cdot a &= a \cdot a' = 1 \\ a'' \cdot a &= a \cdot a'' = 1 \end{aligned}$$

Then:

$$a' = a' \cdot 1 = a' \cdot (a \cdot a'') = (a' \cdot a) \cdot a'' = 1 \cdot a'' = a''$$

1.1.12 Inverse Notation and Operations

Since inverses are unique:

- The additive inverse of a is denoted $(-a)$
- The multiplicative inverse of a is denoted a^{-1}
- Subtraction: $a - b := a + (-b)$
- Division: when $b \neq 0, a/b := a \cdot b^{-1}$

1.2 Consequences

Lecture 05: Consequences

1.2.1 Properties of Zero

Proposition 1.2.1 (Zero Property). Let F be a field. Then $\forall a \in F, a \cdot 0 = 0$.

Proof. Let $a \in F$. Then:

$$\begin{aligned} a \cdot \tilde{0} &= a \cdot \tilde{0} + \tilde{0} \\ &= a \cdot \tilde{0} + (a \cdot \tilde{0} + (- (a \cdot \tilde{0}))) = (a \cdot \tilde{0} + a \cdot \tilde{0}) + (- (a \cdot \tilde{0})) \\ &= a \cdot (\tilde{0} + \tilde{0}) + (- (a \cdot \tilde{0})) = a \cdot \tilde{0} + (- (a \cdot \tilde{0})) = \tilde{0} \end{aligned}$$

■

Proposition 1.2.2 (Zero Product Property). Let F be a field. Then $\forall a, b \in F$:

$$a \cdot b = 0 \Rightarrow a = 0 \text{ or } b = 0$$

Proof. Either $a = 0$ or $a \neq 0$.

If $a = 0$, then the conclusion is satisfied.

If $a \neq 0$, then by field axiom (vii), $\exists a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$. Therefore:

$$b = b \cdot 1 = b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1} = 0 \cdot a^{-1} = 0$$

■

Proposition 1.2.3 (Zero Non-Invertible). Let F be a field. Then $0 \in F$ is not invertible with respect to multiplication.

Proof. Suppose, for contradiction, that $\exists a \in F$ such that $a \cdot 0 = 1$.

Then by Proposition 1.2.1, $1 = a \cdot 0 = 0$.

But by field axiom (viii), $0 \neq 1$, which is a contradiction. ■

1.2.2 Modular Arithmetic

The integers modulo m is the set $\mathbb{Z}_m = \{0, 1, 2, \dots, m-1\}$ with addition and multiplication defined by taking remainders after division by m .

Example. $\mathbb{Z}_2 = \{0, 1\}$ with operation tables:

+	0	1
0	0	1
1	1	0

·	0	1
0	0	0
1	0	1

This is a field.

Example. $\mathbb{Z}_3 = \{0, 1, 2\}$ with operation tables:

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

·	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

This is a field.

Example. $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ is not a field because $2 \cdot 2 = 0$ but $2 \neq 0$, violating the zero product property.

Note. These facts are important in applications but not central to linear algebra

- When m is prime, $\mathbb{Z}_m$ is a field
- When m is not prime, $\mathbb{Z}_m$ is not a field (by the zero product property) (doesn't mean no field with m members isn't possible)

1.2.3 Operations Involving Inverses

Proposition 1.2.4 (Inverse Properties). Let F be a field and let $a, b \in F$. Then:

- $-0 = 0$
- $1^{-1} = 1$
- $-(-a) = a$
- $(a^{-1})^{-1} = a$ (when $a \neq 0$)
- $(-1) \cdot a = -a$
- $(-a) \cdot b = -(a \cdot b)$
- $(-a) \cdot b = a \cdot (-b)$
- $(-a) \cdot (-b) = a \cdot b$

Proof.

- Since $0 + 0 = 0$, we have $-0 = 0$ by uniqueness of additive inverses.
- Since $1 \cdot 1 = 1$, we have $1^{-1} = 1$ by uniqueness of multiplicative inverses.
- Since $(-a) + a = 0$, we have $-(-a) = a$ by uniqueness of additive inverses.
- Since $a^{-1} \cdot a = 1$, we have $(a^{-1})^{-1} = a$ by uniqueness of multiplicative inverses.
- $a + ((-1) \cdot a) = 1 \cdot a + ((-1) \cdot a) = (1 + (-1)) \cdot a = 0 \cdot a = 0$
- $(-a) \cdot b = ((-1) \cdot a) \cdot b = (-1) \cdot (a \cdot b) = -(a \cdot b)$
- $(-a) \cdot b = ((-1) \cdot a) \cdot b = (-1) \cdot (a \cdot b) = a \cdot ((-1) \cdot b) = a \cdot (-b)$
- $(-a) \cdot (-b) = ((-1) \cdot a) \cdot ((-1) \cdot b) = (-1) \cdot (-1) \cdot a \cdot b$
Since $(-1) \cdot (-1) = (-1)^{-1} \cdot (-1) = 1$, we get: $(-a) \cdot (-b) = 1 \cdot a \cdot b = a \cdot b$

■

1.3 Numbers

Note. The essential properties of natural numbers relate to counting and ordering. Other properties (like primality) are nonessential but useful for examples.

Example. The set $\{0, 1, 2, \dots, p-1\}$ with addition and multiplication modulo p is a field if and only if p is prime.

Definition (Integers). Starting with $\mathbb{N}$ and adding zero and additive inverses:

$$\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-x : x \in \mathbb{N}\}$$

Definition (Rationals). Taking ordered pairs of integers:

$$\mathbb{Q} = \{(a, b) : a, b \in \mathbb{Z}, b \neq 0\}$$

Note. Algorithms for addition, multiplication, and ordering extend from $\mathbb{N}$ to $\mathbb{Z}$ to $\mathbb{Q}$. The rigorous justification is handled in real analysis.

1.3.3 Irrational Numbers

The Pythagorean theorem reveals distances that cannot be expressed as ratios of integers.

Example (Pythagoras of Samos). Two unit-length sticks at right angles create a distance $\sqrt{2}$ between their free ends. No ratio of integers accurately describes this distance

Note. The existence of irrational numbers has been known since antiquity. Their rigorous construction (e.g., Dedekind cuts) is handled in real analysis.

Definition (Real Numbers). We accept $\mathbb{R}$ as the union of rational and irrational numbers, using geometric intuition of "length" and the number line.

1.3.4 Real Numbers

Definition 1.3.1 (Complex Numbers). We denote by $\mathbb{C}$ the set of ordered pairs of real numbers:

$$\mathbb{C} = \{(x, y) : x, y \in \mathbb{R}\}$$

with operations:

Equality: $(x, y) = (x', y') \Leftrightarrow x = x' \wedge y = y'$

Addition: $(x, y) + (x', y') = (x + x', y + y')$

Multiplication: $(x, y)(x', y') = (xx' - yy', xy' + x'y)$

Proposition 1.3.2 (Complex Field). The set $\mathbb{C}$ with addition and multiplication as defined is a field.

Proof. We verify each field axiom:

- (i) **Closure:** Since $\mathbb{R}$ is closed under addition and multiplication, both $(x + x', y + y')$ and $(xx' - yy', xy' + x'y)$ belong to $\mathbb{C}$.

- (ii) **Associativity:** For addition:

$$\begin{aligned} [(x, y) + (x', y')] + (x'', y'') &= (x + x', y + y') + (x'', y'') \\ &= ((x + x') + x'', (y + y') + y'') = (x + (x' + x''), y + (y' + y'')) \\ &= (x, y) + [(x', y') + (x'', y'')] \end{aligned}$$

For multiplication:

$$\begin{aligned} [(x, y)(x', y')](x'', y'') &= (xx' - yy', xy' + x'y)(x'', y'') \\ &= ((xx' - yy')x'' - (xy' + x'y)y'', (xx' - yy')y'' + (xy' + x'y)x'') \\ &= (x(x'x'' - y'y'') - y(x'y'' + y'x''), x(x'y'' + y'x'') + y(x'x'' - y'y'')) \\ &= (x, y)[(x', y')(x'', y'')] \end{aligned}$$

- (iii) **Commutativity:** For addition:

$$(x, y) + (x', y') = (x + x', y + y') = (x' + x, y' + y) = (x', y') + (x, y)$$

For multiplication:

$$(x, y)(x', y') = (xx' - yy', xy' + x'y) = (x'x - y'y, x'y + y'x) = (x', y')(x, y)$$

- (iv) **Distributivity:**

$$\begin{aligned} (x, y)[(x', y') + (x'', y'')] &= (x, y)(x' + x'', y' + y'') \\ &= (x(x' + x'') - y(y' + y''), x(y' + y'') + y(x' + x'')) \\ &= (xx' - yy' + xx'' - yy'', xy' + yx' + xy'' + yx'') \\ &= (x, y)(x', y') + (x, y)(x'', y'') \end{aligned}$$

- (v) **Neutral elements:** Let $\tilde{0} = (0, 0)$ and $\tilde{1} = (1, 0)$. Then:

$$\begin{aligned} (x, y) + (0, 0) &= (x, y) \\ (x, y)(1, 0) &= (x \cdot 1 - y \cdot 0, x \cdot 0 + y \cdot 1) = (x, y) \end{aligned}$$

- (vi) **Additive inverses:** Let $-(x, y) = (-x, -y)$. Then:

$$(x, y) + (-x, -y) = (0, 0)$$

- (vii) **Multiplicative inverses:** For $(x, y) \neq (0, 0)$, let:

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

Then $(x, y) \cdot (x, y)^{-1} = (1, 0)$.

- (viii) **Distinct neutrals:** Since $0 \neq 1$ in $\mathbb{R}$, we have $(0, 0) \neq (1, 0)$.

■

1.3.6 Real Numbers as Complex Subset

The mapping $x \mapsto (x, 0)$ identifies $\mathbb{R}$ with the subset $\{(x, 0) : x \in \mathbb{R}\} \subset \mathbb{C}$.

This preserves operations:

$$x + y \mapsto (x + y, 0) = (x, 0) + (y, 0) \quad (1.1)$$

$$xy \mapsto (xy, 0) = (x, 0)(y, 0) \quad (1.2)$$

1.3.7 The Imaginary Unit

Definition (Imaginary Unit). We define $i = (0, 1)$, which satisfies $i^2 = -1$:

$$i \cdot i = (0, 1)(0, 1) = (0 - 1, 0 + 0) = (-1, 0)$$

Notation. Every complex number can be written uniquely as:

$$(x, y) = (x, 0) + (0, y) = x + yi$$

Definition 1.3.3 (Real and Imaginary Parts). For $z = x + yi \in \mathbb{C}$:

$$\Re(z) = x \quad (\text{real part}) \quad (1.3)$$

$$\Im(z) = y \quad (\text{imaginary part}) \quad (1.4)$$

1.3.8 Geometric Interpretation

Complex numbers correspond to points in the Cartesian plane, with the real part as x -coordinate and imaginary part as y -coordinate.

Polar Form

Definition 1.3.4 (Absolute Value and Argument). For $z = x + yi \in \mathbb{C}$:

$$|z| = \sqrt{x^2 + y^2} \quad (\text{absolute value}) \quad (1.5)$$

$$\arg(z) = \arctan\left(\frac{y}{x}\right) \quad (\text{argument, with correct quadrant}) \quad (1.6)$$

For complex numbers $z = |z|(\cos \theta + i \sin \theta)$ and $z' = |z'|(\cos \theta' + i \sin \theta')$:

$$zz' = |z||z'|[\cos(\theta + \theta') + i \sin(\theta + \theta')]$$

The absolute value of the product equals the product of absolute values, and the argument of the product equals the sum of arguments.

1.3.9 Complex Conjugate

Definition 1.3.5 (Complex Conjugate). For $z = x + yi \in \mathbb{C}$, the complex conjugate is:

$$\bar{z} = x - yi$$

Theorem 1.3.6 (Conjugate Properties). For $z, z_1, z_2 \in \mathbb{C}$:

- (i) $\bar{\bar{z}} = z$
- (ii) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
- (iii) $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$
- (iv) $z\bar{z} = |z|^2$
- (v) $z = \bar{z} \Leftrightarrow z \in \mathbb{R}$

1.3.10 Division

For $z, z' \in \mathbb{C}$ with $z' \neq 0$:

$$\frac{z}{z'} = \frac{z\bar{z'}}{z'z'} = \frac{z\bar{z'}}{|z'|^2} = \frac{xx' + yy'}{x'^2 + y'^2} + i \frac{x'y - y'x}{x'^2 + y'^2}$$

Note. Elementary functions (exponential, trigonometric, logarithmic) extend to $\mathbb{C}$ while preserving many properties. The exponential function leads to what many consider the most beautiful equation in mathematics: Euler's identity, $e^{i\pi} + 1 = 0$, while considering functions brings us to complex analysis.

Chapter 2

Systems of Linear Equations

Lecture 07: A Linear Equation Over a Field

2.1 A Linear Equation Over a Field

Proposition 2.1.1. Let F be a field.

- $\forall a, b, c \in F$

$$a + b = a + c \Rightarrow b = c$$
- $\forall a \neq \tilde{0} \in F$ and $\forall b, c \in F$

$$ab = ac \Rightarrow b = c$$

Proof. • $\exists (-a) \in F$ such that $(-a) + a = \tilde{0}$. Then

$$\begin{aligned} a + b = a + c &\Rightarrow (-a) + (a + b) = (-a) + (a + c) \\ \Rightarrow ((-a) + a) + b &= ((-a) + a) + c \Rightarrow \tilde{0} + b = \tilde{0} + c \Rightarrow b = c \end{aligned}$$

- $\exists a^{-1} \in F$ such that $a^{-1}a = \tilde{1}$. Then

$$\begin{aligned} ab = ac &\Rightarrow a^{-1}(ab) = a^{-1}(ac) \\ \Rightarrow (a^{-1}a)b &= (a^{-1}a)c \Rightarrow \tilde{1}b = \tilde{1}c \Rightarrow b = c \end{aligned}$$

Definition 2.1.2 (Standard Form Linear Equation). A **Standard form Linear Equation in n unknowns over a field F** is an expression of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

in which the $x_1, \dots, x_n \in F$ are the **unknowns**, the $a_1, \dots, a_n \in F$ are the **coefficients of the unknowns**, and $b \in F$ is the **free coefficient**.

Definition (Linear Equation). An expression that represents the same condition as some standard form linear equation

Note.

- We often drop an explicit 1 or (-1) as a coefficient of an unknown.
- When the number of unknowns is small, it is sometimes convenient to use a, b, c, x, y, z instead.

Definition 2.1.3 (n -tuple). A finite list of n items, not necessarily distinct, appearing in a particular order.

Note. For 2-tuples and 3-tuples, we often use the terms **pair** and **triple** respectively.

Notation.

- Write components of an n -tuple in parentheses, separated by commas.
- We name an n -tuple with roman, lowercase, underlined letters (e.g., $\underline{a} = (a_1, a_2, \dots, a_n)$).

Note. In print, **a** is more common, but harder when writing.

Definition 2.1.4 (F^n). Let F be a field. The set of all n -tuples whose components are elements of F is denoted by F^n :

$$F^n = \{(a_1, a_2, \dots, a_n) : \forall i \in \{1, 2, \dots, n\}, a_i \in F\}$$

Definition 2.1.5 (Solution). Let $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ be a linear equation over a field F , the n -tuple

$$(c_1, c_2, \dots, c_n)$$

is a **solution** of the equation if this equality is true:

$$a_1c_1 + a_2c_2 + \dots + a_nc_n = b$$

To “solve” an equation with n unknowns over a field F means to characterize precisely the subset of F^n of solutions of that equation.

Example. Let us solve this equation over $\mathbb{Q}$

$$2x - 3y = 7$$

Using the uniqueness of additive and multiplicative inverses in $\mathbb{Q}$, we deduce

$$2x - 3y = 7 \Leftrightarrow -3y = 7 - 2x \Leftrightarrow 3y = 2x - 7 \Leftrightarrow y = \frac{2}{3}x - \frac{7}{3}$$

Therefore, for any $t \in \mathbb{Q}$, the 2-tuple $(t, \frac{2}{3}t - \frac{7}{3})$ is a solution.

We can express the set of solutions, S in set builder notation:

$$S = \left\{ \left(t, \frac{2}{3}t - \frac{7}{3} \right) : t \in \mathbb{Q} \right\}$$

Proposition 2.1.6. Consider the linear equation $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ over a field F .

- If $a_i = \tilde{0} = b, \forall i \in \{1, \dots, n\}$ then every n -tuple in F^n is a solution. The solution set is F^n .
- If $a_i = \tilde{0}, \forall i \in \{1, \dots, n\}$ and $b \neq \tilde{0}$ then no n -tuple in F^n is a solution. The solution set is $\emptyset$.
- If at least one of the $a_i \neq \tilde{0}$ then there exists a solution.

Proof. • By Proposition 1.2.1, for any $(x_1, \dots, x_n) \in F^n$,

$$\tilde{0}x_1 + \tilde{0}x_2 + \dots + \tilde{0}x_n = \tilde{0} = b$$

- By Proposition 1.2.1, for any $(x_1, \dots, x_n) \in F^n$,

$$\tilde{0}x_1 + \tilde{0}x_2 + \dots + \tilde{0}x_n = \tilde{0} \neq b$$

- If $a_i \neq \tilde{0}$ then a_i^{-1} exists and, for any $(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in F^{n-1}$, the n -tuple

$$(x_1, \dots, x_{i-1}, a_i^{-1}(b - a_1x_1 - \dots - a_{i-1}x_{i-1} - a_{i+1}x_{i+1} - \dots - a_nx_n), x_{i+1}, \dots, x_n)$$

is a solution. ■

2.2 A System of Linear Equations

Lecture 08: A System of Linear Equations

2.2.1 System Definition

Definition 2.2.1 (Standard Form Linear System). A **standard form linear system of order $m \times n$** over a field F is a collection of standard form linear equations:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

where the a_{ij} and b_i (for $1 \leq i \leq m, 1 \leq j \leq n$) are scalars of F , called the **coefficients of the system**. The a_{ij} are the **coefficients of the unknowns** and the b_i are the **free coefficients**.

Definition 2.2.2 (Solution of a System). An n -tuple $\underline{c} = (c_1, \dots, c_n) \in F^n$ is a **solution** of an $m \times n$ linear system if it is a solution of each of the m equations of the system.

2.2.2 Consistency

The solution set of a linear system is the intersection of the individual solution sets of each equation.

Definition 2.2.3 (Consistency). If the solution set of a system of linear equations is empty, the system is called **inconsistent**. If the solution set contains at least one solution, the system is called **consistent**.

Note. If any equation in the system has the form:

$$0x_1 + 0x_2 + \dots + 0x_n = b \neq 0$$

then the system is inconsistent. Later, we'll call this a “type-B row”

Note. The algorithm for solving linear systems will transform any inconsistent system to one that is obviously inconsistent.

2.2.3 Redundancy

Definition 2.2.4 (Redundancy). A linear system is called **redundant** if one or more of the equations can be dropped without changing the solution set.

Note. The algorithm for solving linear systems will transform any redundant system to one that is obviously redundant.

2.2.4 Homogeneous Systems

Definition 2.2.5 (Homogeneous System). A linear system is called **homogeneous** if all its free coefficients are zeros. If at least one free coefficient is non-zero, the system is called **non-homogeneous**.

Proposition 2.2.6 (Homogeneous Consistency). Every homogeneous system is consistent.

Proof. Every homogeneous system has at least one solution, namely the n -tuple of zeros: $\underline{0} = (0, 0, \dots, 0)$ ■

Definition 2.2.7 (Trivial Solution). The solution $\underline{0} = (0, 0, \dots, 0)$ is called the **trivial solution** of a homogeneous system.

Note. A homogeneous system may or may not have other solutions besides the trivial solution.

There are no non-homogenous systems where $(\vec{0}, \vec{0}, \vec{0})$ is a solution.

Lecture 09: Equivalent Systems

2.3 Equivalent Systems

Definition 2.3.1 (Equivalent). Two linear systems are called **equivalent** if their solution sets are identical.

Definition 2.3.2 (Elementary Change). An **elementary change** to a linear system is one of the follow changes:

1. Swapping the position of two equations
2. Multiplying every coefficient in one equation by a non-zero scalar
3. Adding the same multiple of the coefficients of one equation to the respective coefficients of another equation

Proposition 2.3.3. Performing an elementary change on a linear system does not change the system's solution set.

Proof. Type (i) Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \leftrightarrow R_j$.

The system includes the same equations before and after the change, so clearly $S = S'$. ■

Proof. Type (ii)

Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \rightarrow tR_i$, where $t \neq 0 \in F$.

All equations except the i -th remain unchanged. For the i -th equation:

Suppose the n -tuple $c = (c_1, c_2, \dots, c_n) \in S$, then

$$a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n = b_i$$

Therefore,

$$ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n = t(a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n) = tb_i$$

Therefore c is also a solution of the i -th equation in the changed system, and $c \in S'$.

Conversely, suppose $c \in S'$, then

$$ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n = tb_i$$

Therefore, because $t \neq 0$,

$$\begin{aligned} a_{i1}c_1 + \dots + a_{in}c_n &= t^{-1}(ta_{i1}c_1 + ta_{i2}c_2 + \dots + ta_{in}c_n) \\ &= t^{-1}(tb_i) = b_i \end{aligned}$$

Therefore, c is also a solution of the i -th equation in the original system, and $c \in S$. Since $S \subseteq S'$ and $S' \subseteq S$, therefore $S = S'$. ■

Proof. Type (iii)

Let S be the solution set of a linear system and let S' be the solution set of the system following the elementary change $R_i \rightarrow R_i + tR_j$ for $t \in F$.

All equations except the i -th remain unchanged. For the i -th equation:

Suppose the n -tuple $c = (c_1, \dots, c_n) \in S$ is a solution of the original system, then

$$a_{i1}c_1 + a_{i2}c_2 + \dots + a_{in}c_n = b_i$$

and

$$a_{j1}c_1 + a_{j2}c_2 + \dots + a_{jn}c_n = b_j$$

Therefore,

$$\begin{aligned} (ta_{j1} + a_{i1})c_1 + \dots + (ta_{jn} + a_{in})c_n \\ &= ta_{j1}c_1 + \dots + ta_{jn}c_n + a_{i1}c_1 + \dots + a_{in}c_n \\ &= t(a_{j1}c_1 + \dots + a_{jn}c_n) + (a_{i1}c_1 + \dots + a_{in}c_n) \\ &= tb_j + b_i \end{aligned}$$

This shows that c is also a solution of the i -th equation in the changed system and therefore $c \in S'$ and $S \subseteq S'$.

Conversely, suppose $c \in S'$ is a solution of the changed system, then

$$(a_{i1} + ta_{j1})c_1 + (a_{i2} + ta_{j2})c_2 + \dots + (a_{in} + ta_{jn})c_n = b_i + tb_j$$

and

$$a_{j1}c_1 + a_{j2}c_2 + \dots + a_{jn}c_n = b_j$$

Therefore,

$$\begin{aligned} a_{i1}c_1 + \dots + a_{in}c_n \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) + 0 \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) - ((a_{i1} + ta_{j1})c_1 + \dots + (a_{in} + ta_{jn})c_n) + (b_i + tb_j) \\ &= (a_{i1}c_1 + \dots + a_{in}c_n) - (a_{i1}c_1 + \dots + a_{in}c_n) + b_i - t(a_{j1}c_1 + \dots + a_{jn}c_n) + \\ &= 0 + b_i - tb_j + tb_j \\ &= b_i \end{aligned}$$

This shows that c is also a solution of the i -th equation in the original system and therefore $c \in S$ and $S' \subseteq S$.

Therefore, since $S' \subseteq S$ and $S \subseteq S'$, we have $S = S'$. ■

Theorem 2.3.4. A **finite sequence of elementary changes** on a linear system leads to an **equivalent** system.

Proof. Let S be the solution set of a linear system and S' be the solution set of the system after performing a finite sequence of elementary changes.

By Proposition 2.3.3, any single elementary change leads to an equivalent system.

Equivalency is transitive, because equality of sets is transitive. ■

Lecture 10: Gaussian Elimination

2.4 Gaussian Elimination with Back Substitution

2.4.1 Coefficient Matrix

Definition 2.4.1 (Matrix). A **matrix** is a rectangular table of scalars from a field.

The number of rows and columns is called the **order** of the matrix. A matrix of order $m \times n$ matrix has m rows and n columns. Individual scalar values are called the **entries** of the matrix.

Notation. We usually name matrices with uppercase Roman letters and matrix elements have the corresponding lowercase letter. For example:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Definition (Augmented Coefficient Matrix). Any linear system of m equations with n unknowns is represented fully by an $m \times (n + 1)$ matrix, called the **augmented coefficient matrix**:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

Definition 2.4.2 (Row Equivalent). Let A and B be matrices of the same order. A is **row equivalent** to B if we can change A into B by a finite sequence of elementary row operations.

Note. Row equivalence is a valid equivalence relation because it is reflexive, symmetric, and transitive.

Definition 2.4.3 (Pivot). A matrix row containing all zeros is called a **null row**. In any row that is not a null row, the leftmost non-zero entry is called the **leading coefficient** or the **pivot**.

2.4.2 Row Echelon Form

Definition 2.4.4 (REF). A matrix is said to be in **row echelon form** (REF) if:

1. In every row that is not a null row the leading coefficient is to the right of all the leading coefficients of the rows above it (if any)
2. All null rows, if any exist, are below all non-null rows

Definition 2.4.5 (eliminated system). A linear system whose augmented coefficient matrix is in row echelon form

Definition 2.4.6 (Restricted and Free Variables). In an eliminated linear system, x_j is called a **restricted variable** if there is a pivot in the j -th column of its coefficient matrix. Otherwise, it is called a **free variable**.

Note. In a solution set of an eliminated system, the restricted variables are specified in terms of the free variables. Each free variable in the general solution n -tuple is represented by a free parameter.

2.4.3 Gaussian Elimination

Given a matrix order of $m \times n$

1. If all entries are zero, the matrix is already in REF. Stop
2. If the matrix has only one row, it is already in REF. Stop
3. Identify the first (leftmost) column containing a non-zero entry, column j
4. If the first (topmost) entry in that column is zero, swap with column with non-zero entry in column j
5. If there are any non-zero entries left in column j below the topmost row, add multiples of the topmost row to the rows of those entries to make them zero
6. Ignore the topmost row and repeat steps 1 – 5 on the remaining sub-matrix

Note. This algorithm works on any field

Lecture 11: Gauss-Jordan Elimination

2.5 Gauss-Jordan Elimination

Definition 2.5.1 (RREF). A matrix is in **reduced row echelon form** (RREF) if it is in REF and:

1. All leading coefficients are $\bar{1}$
2. In a column that includes a leading coefficient, all other coefficients are $\bar{0}$

Note. This definition applies in every field

Given a matrix of order $m \times n$

1. Consider row $i = 1$
2. Identify the first (leftmost) column containing a non-zero entry on or below row i , column j

3. If entry $a_{ij} = 0$, swap row i with any row $k > i$ that doesn't have a zero in column j
4. Now that $a_{ij} \neq 0$, multiply row i by a_{ij}^{-1}
5. If there are any non-zero entries left in column j , add multiples of row i to the rows of those entries to make them zero.
6. Increase $i \rightarrow i + 1$ and repeat steps 2 – 5

However, for solving a system of linear equations, RREF is not worth it (REF is faster and easy enough ($\frac{1}{2}O(n^3)$ vs $\frac{1}{3}O(n^3) + \frac{1}{2}O(n^2)$))

Lecture 12: How Many Solutions Are There?

2.6 System Solution Analysis

2.6.1 Consistency

Theorem 2.6.1 (Consistency Condition). A linear system is consistent if and only if its augmented coefficient matrix is row equivalent to a matrix in reduced row echelon form that does not contain a type-B row.

Proof. Form the augmented coefficient matrix of the system and run the Gauss-Jordan elimination algorithm.

- All operations during the algorithm are elementary row operations
- All matrices (initial, final, and intermediate) are row equivalent to the original matrix
- All linear systems represented by these matrices are equivalent
- If at any point a type-B row appears, the system is inconsistent
- If the algorithm completes, the output is a RREF matrix without a type-B row
- This matrix represents a system equivalent to the original system
- A solution is available by back substitution

2.6.2 Number of Solutions

Let A be a linear system with n unknowns over a field F , and suppose the augmented coefficient matrix is row equivalent to a RREF that does not contain a type-B row.

1. If the RREF matrix has exactly n non-zero rows, then the system has a **unique solution**
2. If the RREF matrix has $k < n$ non-zero rows, then there are $n - k$ free variables:
 - If field F has infinitely many scalars, there are **infinitely many solutions**
 - If field F has N different scalars, there are N^{n-k} solutions

Note. The RREF matrix has at most n non-zero rows because each non-zero row has a pivot in one of the first n columns, and everything else in a pivot column is zero.

2.6.3 Homogeneous Systems

Note. A homogeneous system is always consistent because the trivial solution always exists.

Theorem 2.6.2 (Non-trivial Solutions). A homogeneous system with more unknowns than equations has a non-trivial solution.

Proof. Let A be a homogeneous system with m equations in n unknowns where $m < n$.

- The RREF of the coefficient matrix has no type-B rows (system is homogeneous)
- There are at most m pivots in the RREF matrix (at most one per row)
- Therefore, there are at most m restricted variables
- Therefore, there is at least one free variable since $n - m > 0$
- Any choice of $t \in F$ for the free variable leads to a solution that is an n -tuple with t as one component
- Including the choice $t = 1$
- Since $1 \neq 0$ by axiom viii in Definition 1.1.3, there exists a non-trivial solution

Lecture 13: Square Systems and Square Matrices

2.7 Square Matrices

An $n \times n$ matrix is called a **square matrix**. A linear system with n equations in n unknowns has a square coefficient matrix.

Notation. To avoid confusion with coefficient matrices, we will use the following terminology:

- **Augmented coefficient matrix:** includes the constants column $[A|b]$
- **Not-augmented coefficient matrix:** contains only the coefficients of the unknowns

In a square matrix of order n , the **main diagonal** is the diagonal from upper left to lower right, consisting of entries a_{ii} for $i \in \{1, 2, \dots, n\}$.

Definition 2.7.1 (Identity Matrix). The square matrix of order n whose main diagonal entries are equal to $\bar{1}$ and all other entries are equal to $\bar{0}$ is called the **identity matrix of order n** , denoted I_n .

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

The element in row i , column j of I_n equals δ_{ij} (Kronecker delta).

2.7.1 RREF and Identity Matrix

Theorem 2.7.2 (RREF Square Matrix). If A is an RREF square matrix of order n with a pivot in every row, then $A = I_n$.

Proof. Let A be as described. Since there's a pivot in every row, there are n pivots total.

In RREF, every pivot is in a different column. Suppose the pivot of the i -th row is in the j -th column.

- There are $i - 1$ rows above the i -th row, whose pivots must be in columns strictly to the left of column j , so $i - 1 \leq j - 1$
- There are $n - i$ rows below the i -th row, whose pivots must be in columns strictly to the right of column j , so $n - i \leq n - j$

Therefore $i \leq j$ and $j \leq i$, which implies $i = j$.

The pivots of A are therefore a_{ii} for $i \in \{1, \dots, n\}$. In RREF:

- All pivots equal $\bar{1}$
- In every pivot column, all non-pivot entries equal $\bar{0}$

Therefore $a_{ij} = \delta_{ij}$, which is the definition of I_n . ■

2.7.2 Unique Solutions for Square Systems

Lemma 2.7.3. If A and B are row equivalent, and we delete column j from each to make A' and B' respectively, then A' and B' are also row equivalent.

Proof. The same sequence of row operations that leads from A to B also leads from A' to B' . ■

Theorem 2.7.4 (Unique Solution Test). A linear system of n equations in n unknowns has a unique solution if and only if its not-augmented coefficient matrix is row equivalent to I_n .

Proof. Let $A = [A'|b_\uparrow]$ be the augmented coefficient matrix, where A' is the not-augmented coefficient matrix.

($\Rightarrow$): Suppose A' is row equivalent to I_n .

Apply the same sequence of operations to $A = [A'|b_\uparrow]$ making $[I_n|c_\uparrow]$

This represents the system: $x_1 = c_1, x_2 = c_2, \dots, x_n = c_n$

This system has the unique solution $\underline{c} = (c_1, c_2, \dots, c_n)$

($\Leftarrow$): Suppose the system has a unique solution.

Apply row operations to get RREF matrix $C = [C'|c_\uparrow]$

Since the original system has one solution, so does the equivalent RREF system

Therefore, there are no free variables, so C has n pivots in the first n columns

Therefore C' has a pivot in every row and every column

By Theorem 2.7.2, $C' = I_n$

By Lemma 2.7.3, A' is row equivalent to I_n ■

Corollary 2.7.5. For a square matrix A : if one linear system with coefficient matrix A has a unique solution, then **all** linear systems with coefficient matrix A have unique solutions.

Equivalently: A square system $[A|b]$ has a unique solution if and only if the corresponding homogeneous system $[A|0]$ has no non-trivial solution.

2.7.3 Square Homogeneous Systems

Theorem 2.7.6 (Square Homogeneous Dichotomy). For the not-augmented coefficient matrix of a homogeneous system of n equations in n unknowns, exactly one of the following is true:

1. It is row equivalent to a matrix with a row of zeros $\Leftrightarrow$ the system has a non-trivial solution
2. It is row equivalent to $I_n \Leftrightarrow$ the system has no non-trivial solution

Proof. Proof of (1):

- ($\Rightarrow$) Suppose A is row equivalent to a matrix B with a row of zeros
- WLOG, assume the zero row is the last row (swap if necessary)
- Apply row operations to get RREF matrix C (the zero row remains)
- A square RREF matrix with a zero row has fewer than n pivots
- Therefore the system has at least one free variable
- Therefore the homogeneous system has a non-trivial solution
- ($\Leftarrow$) If A has a non-trivial solution, it doesn't have a unique solution
- By Theorem 2.7.4, A is not row equivalent to I_n
- By Theorem 2.7.2, the RREF of A doesn't have a pivot in every row
- Therefore it has a row of zeros

Proof of (2):

- ($\Rightarrow$) If A is row equivalent to I_n , the system has a unique solution
- Since it's homogeneous, that unique solution is the trivial solution
- ($\Leftarrow$) If the system has no non-trivial solution, by part (1) it's not row equivalent to a matrix with zero rows
- Therefore the RREF of A has no zero rows
- By Theorem 2.7.2, the RREF of A is I_n ■